

Premankur Banerjee

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Ph.D. researcher specializing in Haptics, VR/AR, HCI and Robotics, with over 7 years of hands-on experience building end-to-end multisensory, immersive, and interactive systems. From hardware design and real-time control algorithms to software integration and extensive user studies, my work focuses on transforming prototypes into validated, scalable solutions, making bold ideas evolve into innovative prototypes and *impossible* interactions part of everyday technology.

EDUCATION

University of Southern California

Ph.D., Computer Science, GPA: 3.95 / 4.00

August 2023 – Present

Expected Graduation – August 2027

Technical University of Munich

M.Sc. (*with distinction*), Electrical and Computer Engineering, GPA: 1,4 / 1,0

October 2019 – December 2021

University of Calcutta

B. Tech, Electronics and Communications Engineering, GPA: 9.16 / 10.00

August 2015 – May 2019

EXPERIENCE

Graduate Research Assistant, Haptics Robotics and Virtual Interactions (HaRVI) Lab

Thomas Lord Department of Computer Science

University of Southern California, Los Angeles, CA

August 2023 – Present

Surfing in Virtual Reality (Project Lead):

- Developed a [VR surfing simulation](#) integrating a 6-DoF motion platform in Unity 3D, to study the effect of surf therapy on patients, promoting physical and psychological well-being
- Implemented interaction techniques for navigating in VR, via external tracking using Depth Cameras
- Developed control optimization algorithms, real-time state estimation techniques, and optimized multi-DOF motion dynamics for improved force feedback, motion perception and real-time physics control to increase simulation fidelity and user experience
- Designed and engineered force sensor platforms for bi-directional control, enabling maneuverability through center of pressure shifts, improving control accuracy and seamless user interactions in the simulator
- Implemented [multisensory feedback](#) (airflow, vision, motion) to increase immersion and reduce cybersickness, leading to 35% improvement in user comfort and engagement.
- Designed and developed thermal + vibrotactile gloves to simulate water interaction while surfing in VR
- Conducted controlled user studies with over 50 participants

Multi-User Virtual System for Bidirectional, Mediated, Real-Time Social Touch (Project Lead):

- Developed a [haptic-enabled multi-user VR and MR system](#) for real-time, bidirectional social touch, enhancing presence and embodiment scores by 45%
- Designed and built haptic gloves and sleeves with 52 vibrotactile (ERM) actuators in total
- Implemented a WiFi-based cross-device communication protocol, optimized for remote touch interactions with Arduino and Unity, reducing latency to under 100 ms
- Conducted user studies, demonstrating that visual-haptic feedback improved pleasantness and realism of social interactions by 38% when compared to visual-only interactions, leading to higher emotional engagement

Redirected Walking (RDW) in Virtual Reality using Ankle-Based Haptics (Project co-lead):

- Designed a wireless, gait-synchronized ankle-based vibrotactile system for RDW in VR, using 4 ERM actuators and 2 ankle-mounted IMUs to deliver swing-phase haptic cues.
- Conducted user study with 28 participants across 14 RDW conditions, analyzing how vibration side, curvature direction, and gait-velocity-modulated feedback affect curvature detection thresholds, presence, simulator sickness
- Demonstrated that unilateral ankle stimulation meaningfully improved RDW performance, increasing curvature gain thresholds by up to 73.1% and enabled imperceptible redirection along tighter walking arcs, while movement-contingent feedback improved perceived naturalness, presence, and comfort.
- Derived design guidelines for scalable haptic RDW systems, showing that locomotor-state adaptation, spatial asymmetry, and gait-synchronized feedback are key to extending VR walking using lightweight wearable haptics.

Disability-Centered Vibrotactile System (Project co-lead, *in collaboration with* USC School of Cinematic Arts – Media Arts + Practice, [Leonardo Crip Tech Incubator](#), and [Narrative and Emerging Media Program, ASU](#)):

- Built a wireless, modular vibrotactile wearable system with a real-time Unity control and authoring interface (GUI), for remote, multi-actuator haptic composition, recording, and playback.
- Led co-creative, user-centered system design and evaluation with Disabled artists, validating usability, engagement, and affective expressivity through controlled studies and live performance deployment.

**Research Engineer, [Human-Computer Interface Group](#),
Institut des Systèmes Intelligents et de Robotique (ISIR),
Sorbonne Université, Paris, France**

June 2022 – May 2023

- Worked on real-time force feedback control algorithms for a novel encounter-type haptic device delivering >100N of kinesthetic feedback, enabling full-body interactions in room-scale VR environments
- Designed and implemented dynamic VR scenarios integrating body-scale haptic interactions (e.g., pushing, leaning, pulling) along with motion tracking (Optitrack) to enhance physical realism
- Investigated low-cost haptic solutions by leveraging redirected walking and visio-haptic illusions, reducing the need for large-scale actuators while maintaining perceptual realism
- Led experimental validation by conducting user studies

**Master Thesis Student,
German Aerospace Center (DLR), Munich, Germany**

May 2021 – November 2021

Multimodal VR Evaluation Suite for Novel Tactile Wearable Devices:

- Built a tactile device-agnostic [virtual environment](#) for evaluation and comparison of multiple tactile devices
- Designed multiple interaction techniques for tactile modalities such as friction, temperature, texture, and shape
- Conducted user studies with two novel tactile devices to validate the proposed evaluation suite

**Research Intern, Chair of Media Technology (LMT)
Technical University of Munich, Munich, Germany**

November 2019 – April 2021

- Developed an Autoencoder based compression technique to transmit vibrotactile signals over the internet
- Simulated Kinova MOVO Robot in Unity3D by connecting ROS and Unity for [liquid pouring tasks](#)
- Migrated Kinova MOVO for new versions of Gazebo, ROS, and developed autonomous grasping algorithms

PUBLICATIONS

Journal Articles

- **P. Banerjee**, S. Sweidan, C. Yunis, V. H. Cruz, H. Culbertson, [Crip Haptic Jam: Vibrotactile Composition for Disability-Centered Dance](#), **IEEE Transactions on Haptics (ToH) Special Issue: Haptics for Arts**
- **P. Banerjee**, M. Montiel, L. Tomita, O. Means, J. Kutch, H. Culbertson, [The Impact of Airflow and Multisensory Feedback on Immersion and Cybersickness in a VR Surfing Simulation](#), **IEEE Transactions on Visualizations and Computer Graphics (TVCG)**, presented at IEEE VR 2025.

Conference Proceedings and Demos

- **P. Banerjee**, N. Li, J. Wang, H. Culbertson, [Shaping Redirected Walking with Ankle-Based Vibrotactile Feedback](#), *Accepted at ISMAR 2026*.
- **P. Banerjee**, J. Wang, O. Means, H. Culbertson, [SWIMVR: Simulating Water Interaction using Multimodal Vibro-Thermal Rendering](#), **IEEE Haptics Symposium, 2026**.
- **P. Banerjee**, J. Wang, L. Tomita, M. Montiel, H. Culbertson, [Virtual Encounters of the Haptic Kind: Towards a Multi-User VR System for Real-Time Social Touch](#), **2025 IEEE World Haptics Conference (WHC)**.
- **P. Banerjee**, J. Cherin, J. Upadhyay, J. Finley, J. Kutch, and H. Culbertson, [Perception and Control of Surfing in Virtual Reality using a 6-dof Motion Platform](#). In Proc., **EuroHaptics Conference, 2024**.
- H. Singh, L. P. Marcilla, **P. Banerjee**, M. Rothhammer, T. Hulin, [FerroVibe: Towards a Modular Tactile Device](#), **2024 IEEE Conference on Telepresence**.
- **P. Banerjee**, E. Muschter, H. Singh, B. Weber and T. Hulin, [Towards a VR Evaluation Suite for Tactile Displays in Telerobotic Space Missions](#), **2023 IEEE Aerospace Conference**, Big Sky, MT, USA, 2023.

- H. Singh, T. Hulin, P. Banerjee, [Rendering and Displaying Tactile Properties of Virtual Objects via ViESTac and FerroVibe](#), 2023 IEEE World Haptics Conference, WHC 2023, Demos.

Theses

- Banerjee, Premankur (2021) [ViESTac: A Multimodal VR Evaluation Suite for Novel Tactile Devices](#). DLR-Interne Bericht. DLR-IB-RM-OP-2021-214. Master's. Technische Universität München. 104 S.

TEACHING EXPERIENCE

- Teaching Assistant, **DSCI-517: Research Methods and Analysis for User Studies**, Fall 2024, USC
- Teaching Assistant, **EI5052: Time Varying Systems and Computation**, Winter 2021, TUM
- Teaching Assistant, **EI72071: Computational Haptics Laboratory**, Summer 2021, TUM
- Teaching Assistant, **EI7341: Image and Video Compression**, Spring 2021, TUM
- Teaching Assistant, **EI70220: Digital Signal Processing**, Summer 2020, TUM

HONORS, DISTINCTIONS, AND FELLOWSHIPS

- **Fellowships:**
 - [Link Foundation Fellowship](#) in Modeling, Simulation, and Training, \$35,000 (2026–2027).
 - Viterbi School of Engineering Fellowship, University of Southern California, \$42,000 (2023–2024).
 - *Deutschlandstipendium* (German National Scholarship), Technical University of Munich for academic performance and contributions to the scientific community, €7,200 (2020–2021).
- **Awards:**
 - Undergraduate Mentor Award, Thomas Lord Department of Computer Science, Viterbi School of Engineering, University of Southern California (2025).
- **Press and Media Coverage:**
 - Multi-user mediated social touch project featured by [USC news](#), [TikTok/YouTube](#), Arduino [official blog](#)
 - Research on *Surfing in Virtual Reality* featured by [KTLA press](#) and published in [USC Viterbi Magazine](#)
- **Science and Technology Outreach:**
 - Demonstrated *CoVR* at [Fête de la Science](#), contributing to public engagement in science and technology

SKILLS

Technical Skills

- **Programming & Software:** C++, C, C#, Python, MATLAB, Unity 3D, CHAI3D, OpenHaptics, Arduino, Microcontrollers, Blender, FreeCAD, ROS, Gazebo, Tensorflow
- **Haptics, VR, Sensing:** Actuators, Force Sensing and Feedback, Tactile Wearables, Filtering, Temporal Smoothing, Multiplayer VR/AR, Pressure Sensing, Gesture-based Input, Motion Tracking (OptiTrack, Intel RealSense), 6-DoF Motion Platforms, Telepresence
- **Simulation & Robotics:** Physics Modeling, OpenGL, Room-Scale VR, Autonomous Robotics, Inverse Kinematics, IMUs, Sensor Fusion, Wireless Communication Protocols, Teleoperation
- **Machine Learning:** CNNs, Autoencoders, Signal Processing, Interactive ML
- **User-studies & Experimentation:** Experimental Design, User-centric design, Protocol creation, Data Processing, Statistical Analyses (ANOVA, non-parametric tests, mixed models, and others), conducted over 200 user studies

Soft Skills

- **Leadership & Teaching:** Mentored 20 graduate and undergraduate students at USC
- **Collaboration & Communication:** Worked across interdisciplinary teams at USC, TUM, Sorbonne University
- **Project Management:** Led multiple projects, managed hardware-software integration in research environments

ADDITIONAL INFORMATION

- Languages – English (native), Bengali (native), Hindi (native), German (professional), French (elementary)

- Peer Reviewer for IEEE Transactions on Haptics (ToH), IEEE Robotics and Automation Letters (RA-L), Springer Nature – Virtual Reality, IEEE World Haptics, IEEE Haptics Symposium, EuroHaptics, IEEE VR, Perspectives of Communication & Media (PCM), Wisdom Academic Press
- Hobbies – Acting, Filmmaking, VFX design, Reading, Photography, Singing, Film scoring